Aerodynamics

Where drag comes from, what Mach number changes, why nose shape and rocket size matter, and why the fairing leaves early.

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HOW TO USE THIS CHAPTER

- **Print single-sided** and write in the margins.
- Do each example with a calculator first. Air properties are standard-atmosphere values.
- ISRO publishes LVM3's dimensions (43.5 m tall, 4 m body, 5 m fairing) and fairing release (115 km, 217 s). It does not publish drag coefficients or Mach numbers, so those are typical or assumed values and labelled.

By the end of this chapter you can: use the drag equation; compute the speed of sound and Mach number; explain why drag changes around Mach 1; compare nose shapes; and show why big rockets lose proportionally less to drag.

1 The drag equation

NASA Glenn describes drag as the aerodynamic force that resists motion through air. Chapter 3 introduced it. Here is the full form:

$$D = C_d \times \tfrac{1}{2}\rho v^2 \times A$$

ρ = air density (kg/m³) $\quad$ v = speed (m/s)
A = frontal area (m²) $\quad$ C_d = drag coefficient (a shape number)

Double the speed and drag becomes four times larger. Halve the air density and drag halves. C_d packs everything about shape, surface and flow type into one number, found by wind-tunnel or flow-simulation tests. A smooth, pointed rocket has a lower C_d than a flat-nosed one.

ISRO LENS: THE WIDEST PART SETS THE AREA

LVM3 has a 4 m body, but its payload fairing is 5 m in diameter. The frontal area that matters is set by the widest cross-section: the fairing gives $\pi \times 2.5^2 = 19.6$ m², not the 12.6 m² of the 4 m body. That is why Chapter 3's what-if drag was too low, and Example B corrects it.

2 Mach number: speed relative to sound

Sound is a pressure disturbance travelling through the air. Its speed depends only on air temperature:

```
a  =  √( γ R T )          γ = 1.4,  R = 287 J/(kg·K),  T in kelvin
Mach number  M  =  v / a
```

When a rocket is far below the speed of sound (M below about 0.8), air ahead can "hear" it coming and flows smoothly around it. Near and above M = 1, the air cannot get out of the way in time. Pressure piles up in thin, sudden fronts called **shock waves**, and the drag behaves quite differently.

3 Why drag changes near Mach 1

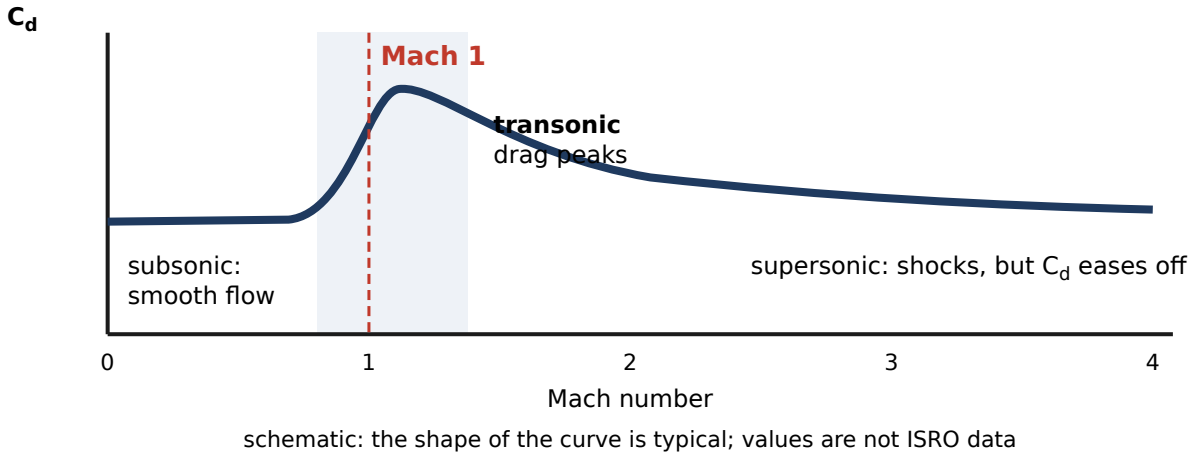


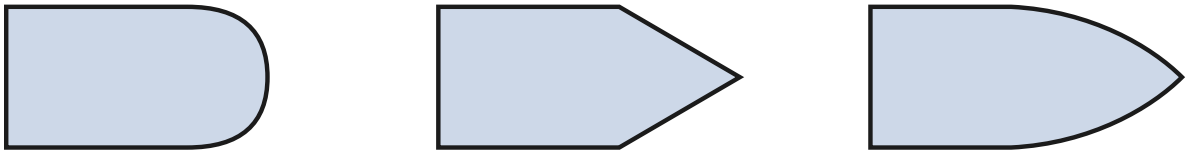
Figure 1. Drag coefficient against Mach number. It climbs steeply through the transonic range, peaks near Mach 1, then falls somewhat. The shaded band is the region designers worry about.

SHOCK WAVES, IN ONE PAGE

A shock wave is a very thin region where air pressure, density and temperature jump almost instantly. A rocket creates shocks at its nose and where its shape changes sharply (shoulders, strap-ons, fins). Shocks cost energy, which shows up as extra drag. A pointed, smoothly curved nose weakens them. That is all you need for this series. NASA's guide develops the mathematics (normal and oblique shock relations, Euler equations); we skip those because the sums add little for understanding a launch.

Rockets deal with this by accelerating through the transonic range while the air is already thinning. Ascent timing and shape together keep the transonic drag peak from taking too many seconds of thrust.

4 Shape: nose and fairing



Blunt
highest drag, roomy

Cone
simple, low drag

Ogive
low drag and good volume

Figure 2. Nose shapes, flying to the right in air coming from the right. Pointed shapes give lower drag at high speed; a curved ogive also leaves room inside for the payload. Most launch-vehicle fairings are ogive-like. Qualitative comparison.

The nose is not the only part. The cylinder behind it, the boosters strapped alongside and the fairing's shoulder where the diameter changes all add drag. Designers also care about **fineness ratio**, length divided by diameter. LVM3's is 43.5 / 4 ≈ 10.9 based on the body, or 8.7 using the 5 m fairing. Long, slender rockets make less drag per unit of volume than short, stubby ones.

5 Why big rockets lose less

Scale a rocket up by a factor L in every direction. Frontal area grows as L^2. Volume, and so mass at the same density, grows as L^3. Drag is proportional to area, and weight to mass, so **drag divided by weight falls as 1/L**. A large rocket carries proportionally more mass behind each square metre of nose, so air slows it less. This is one reason small launchers struggle with drag losses.

6 Worked examples

Example A: speed of sound and Mach number

```
Sea level, 288 K:   a = √(1.4 × 287 × 288.15) ≈ 340 m/s
At 11 km, 216.65 K:  a = √(1.4 × 287 × 216.65) ≈ 295 m/s
Chapter 3's what-if (400 m/s at 11 km):  M = 400 / 295 ≈ 1.36
```

Air is colder aloft, so sound is slower there. The what-if was already supersonic, just above the transonic peak.

Example B: drag with the real frontal area

Same assumptions as Chapter 3 ($\rho = 0.364$ kg/m³, v = 400 m/s, mass 400 t) but use the 5 m fairing, $A = 19.6$ m².

```
q = ½ρv² = ½ × 0.364 × 400² ≈ 29,100 Pa
C_d = 0.3:  D = 0.3 × 29,100 × 19.6 ≈ 171 kN   (4.4% of 3.9 MN weight)
C_d = 0.5:  D = 0.5 × 29,100 × 19.6 ≈ 286 kN   (7.3% of weight)
```

The wider fairing raises drag by 56% over Chapter 3's 4 m estimate (110 kN). A transonic C_d of 0.5 would be worse still. Both C_d values are assumptions.

Example C: drag and speed

```
Same ρ, C_d = 0.3, A = 19.6 m²
v = 100 m/s:  q = 1,820 Pa    D ≈  10.7 kN
v = 200 m/s:  q = 7,280 Pa    D ≈  42.9 kN
v = 400 m/s:  q = 29,100 Pa   D ≈ 171 kN
```



Figure 3. Drag grows with the square of speed (scale 2.4 px per kN).

Doubling the speed quadruples the drag. Four times the speed would give sixteen times the drag at the same density, which is why the rocket needs the air to thin as it accelerates.

Example D: the scale effect

Compare LVM3 with a geometrically similar model one tenth its size, same density and speed.

```
Area:  (1/10)² = 1/100        Mass: (1/10)³ = 1/1000
Drag/weight of the model = (1/100)/(1/1000) = 10 times LVM3's
```

Shrinking the rocket makes drag ten times more important relative to weight.

Example E: why heating is mild at max-Q but matters later

The air in front of the nose is slowed almost to a stop and heated. The stagnation temperature is $T_0 = T \times (1 + 0.2\ M^2)$.

```
T = 216.65 K,  M = 1.36:   T0 = 216.65 × (1 + 0.2 × 1.84) ≈ 296 K
(about 23 °C)
```

At max-Q conditions the nose barely warms above room temperature. But heating climbs steeply with speed, so it becomes a design issue as speed rises. By the time LVM3 is at 115 km, the air is millions of times thinner than at sea level; the fairing's job is done, so ISRO drops its mass.

CHAPTER SUMMARY

- Drag $D = C_d \times \frac{1}{2}\rho v^2 \times A$. It grows with speed squared and with frontal area, and falls as the air thins.
- Speed of sound $a = \sqrt{(\gamma RT)}$ is about 340 m/s at sea level and 295 m/s at 11 km. Mach number = v/a.
- Drag peaks around Mach 1 because of shock waves and falls somewhat above it.
- Pointed or ogive noses and slender bodies cut drag. The widest section (LVM3's 5 m fairing) sets the frontal area.
- Drag/weight falls as 1/L with size: big rockets lose proportionally less.

Next, Chapter 5: stability and control. Why the centre of pressure must sit behind the centre of gravity and how a rocket is steered.

Practice problems

Use $\gamma = 1.4$, $R = 287$ J/(kg·K). Answers are on the last page.

P1. Speed of sound at 216.65 K, and the Mach number at 350 m/s there.

P2. Drag at sea level ($\rho = 1.225$ kg/m³), $v = 100$ m/s, $C_d = 0.3$, $A = 12.6$ m² (4 m body).

P3. By what factor does drag rise if the speed triples at the same air density?

P4. A rocket is scaled to half the length, same shape and density. By what factor does drag/weight change?

P5. In Example B, what is the drag if C_d were 0.4 instead of 0.3 or 0.5? What does the answer tell you about the importance of knowing C_d?

P6. Why does ISRO keep the fairing on until 115 km instead of dropping it at, say, 40 km? Give two reasons.

Glossary

Drag coefficient (C_d)
A shape-dependent number in the drag equation.

Frontal area (A)
The cross-section facing the airflow.

Speed of sound (a)
$\sqrt{(\gamma RT)}$; depends on temperature.

Mach number (M)
Speed divided by the local speed of sound.

Subsonic, transonic, supersonic
Below, near and above Mach 1.

Shock wave
A thin front where pressure, density and temperature jump suddenly.

Ogive
A smooth curved nose shape.

Fineness ratio
Length divided by diameter.

Stagnation temperature
The temperature air reaches when brought to rest: $T(1 + 0.2M^2)$.

Dynamic pressure (q)
$\frac{1}{2}\rho v^2$; the air's push on the vehicle.

Answer key

Fold this page back before you start the problems.

P1. a = √(1.4 × 287 × 216.65) ≈ 295 m/s. M = 350/295 ≈ 1.19.

P2. q = ½ × 1.225 × 100^2 = 6,125 Pa. D = 0.3 × 6,125 × 12.6 ≈ 23.1 kN.

P3. 3^2 = 9 times.

P4. Area ÷ 4, mass ÷ 8, so drag/weight × 2.

P5. D = 0.4 × 29,100 × 19.6 ≈ 229 kN. The range 171 to 286 kN spans a factor of 1.7, so an uncertain C_d makes drag loss uncertain by the same factor. That is why designers test shapes in wind tunnels.

P6. At 40 km there is still some air, and the fairing protects the payload from aerodynamic loads and heating; ISRO's own timeline drops it at 115 km, when the air is thin enough that the shield is no longer needed. Dropping it earlier risks the satellite; holding it much longer carries dead mass the upper stage must keep accelerating.

Sources

Theory: NASA Glenn Research Center, Beginner's Guide to Rockets (drag, Mach number, shock waves, drag coefficient). ISRO: LVM3 launch vehicle page (dimensions, fairing separation at 115 km and 217 s). Standard-atmosphere air properties and the C_d-versus-Mach shape are textbook values. Drag coefficients and the 400 t, 400 m/s, 11 km case are assumptions; ISRO does not publish them.